Answer Scheme

25901158

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECCHE200 - BASIC ANALYTICALAND COSMETIC CHEMISTRY

(2024 ADMISSION ONWARDS)
Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very short answer questions
Answer any 10 questions
Each question carries 2 marks

1. **Explain the advantages of hand sanitizer.**
   Advantages of hand sanitizer, Kills bacteria and viruses; convenient to use without water; helps maintain hand hygiene.
2. **What is the function of preservatives in hand wash formulation?**
   Prevent microbial growth; increase shelf life; maintain product stability.
3. **Difference between acetone and non acetone removers**
   Acetone: fast acting but drying. Non acetone: milder solvents like ethyl acetate; gentler on nails.
4. **Describe the Factors affecting nail growth**
   Nutrition, age, hormonal factors, health condition, and blood circulation.
5. **How are soaps classification of soaps based on TFM**
   Grade 1: >76%; Grade 2: 70–76%; Grade 3: <70%.
6. **What is the active ingredient in clove used in oral care**
   Eugenol – has antibacterial and pain relieving properties.

7. **How Classification of soaps based on pH**
   Neutral soaps, mild alkaline soaps, medicated soaps (generally pH 9–11).
8. **Name any Oils used in soap manufacture**
   Examples: coconut oil, palm oil, olive oil, castor oil, neem oil.

9. **What is the Role of soil minerals in soil composition**
   Soil minerals are essential components that determine the physical and chemical properties of soil. They provide essential nutrients like potassium, calcium, magnesium, and iron required for plant growth. Minerals also influence soil texture, structure, water retention capacity, and fertility. Additionally, they play a key role in maintaining soil pH and supporting microbial activity in the soil.
10. **What you mean by Reverse Osmosis (RO) purification?**
    A water purification process using a semi permeable membrane to remove dissolved salts and impurities.
11. **Write the chemical formula of water and name its constituent elements.**

H2O – elements: hydrogen and oxygen.

12. **Write any two applications of soil pH analysis**

Determines soil fertility; helps select crops; guides fertilizer use.

Part B
Short Answer Questions
Answer any 6 questions
Each question carries 5 marks

13. **Discuss how hand care products help maintain skin health.**

Hand care products such as soaps, hand washes, sanitizers, and moisturizers play an important role in protecting and nourishing the skin.

- **Cleansing action:** Soaps and hand washes remove dirt, oils, and harmful microorganisms, helping to maintain hygiene and prevent infections.
- **Moisturization:** Creams and lotions contain emollients and humectants that keep the skin hydrated and prevent dryness and cracking.
- **Protection:** Some products form a protective barrier on the skin, shielding it from environmental damage such as pollution and chemicals.
- **Antimicrobial effect:** Hand sanitizers and medicated hand washes reduce or kill germs, lowering the risk of disease transmission.
- **Skin nourishment:** Many hand care products include vitamins and natural extracts that improve skin texture and promote healing**. (Each point have one mark)**

14. **Explain the basic steps involved in making a hand wash**

Selection of ingredients:

Choose suitable surfactants (cleansing agents), water, thickening agents, preservatives, fragrance, and color.

Preparation of base solution:

Take distilled water and slowly add the primary surfactant (e.g., SLES) with gentle stirring to avoid foam formation.

Addition of secondary ingredients:

Add co-surfactants, humectants (like glycerin), and thickening agents to improve texture and moisturizing property.

Incorporation of additives:

Add preservatives to prevent microbial growth, followed by fragrance and color as required.

pH adjustment:

Adjust the pH to a skin-friendly range (around 5.5–7) using suitable agents like citric acid.

Mixing and homogenization:

Stir the mixture thoroughly to obtain a uniform and stable formulation.

- **Filling and packaging:**

  Transfer the prepared hand wash into clean containers and seal properly.

  (Each point have one mark)

15. **Describe the meanings and significance of the terms nail acquer and nail enamel.**

Nail Polish:

Nail polish is a cosmetic preparation applied to fingernails or toenails to decorate, protect, and enhance their appearance. It typically consists of film-forming agents (like nitrocellulose), solvents, plasticizers, resins, and color pigments.

Nail Lacquer:

- *Meaning:* Nail lacquer refers to a type of nail polish with a high-gloss, smooth, and durable finish.
- *Significance:* It forms a thin, hard film on the nail surface that dries quickly and provides a shiny, long-lasting coating. It is commonly used for its aesthetic appeal and quick-drying property.

Nail Enamel:

- *Meaning:* Nail enamel is a thicker and more opaque form of nail polish.
- *Significance:* It provides better coverage, stronger adhesion, and enhanced protection to the nail. It often gives a richer color and smoother finish, helping to prevent chipping and damage. **(Each point have 1.5 marks)**
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16. **Discuss the role of humectants, binders, and surfactants used in toothpaste**.

1. Humectants:

Humectants such as glycerin, sorbitol, and propylene glycol help retain moisture in toothpaste. They prevent the paste from drying out and maintain a smooth, soft texture. They also contribute to a pleasant mouthfeel.

2. Binders (Thickening agents):

Binders like carboxymethyl cellulose, gums, or carrageenan help maintain the consistency and stability of toothpaste. They prevent separation of solid and liquid components, ensuring a uniform and stable formulation.

- **3. Surfactants:**
  Surfactants such as sodium lauryl sulfate (SLS) provide foaming action. They help in loosening and removing food particles, plaque, and debris from teeth. Surfactants also aid in the even distribution of toothpaste throughout the mouth.
  (Each point have 1.5 mark)

17. **Discuss the procedure for the preparation of cream soaps.**

Cream soaps are soft, semi-solid soaps with good moisturizing properties. Their preparation involves the following steps:

1. **Selection of raw materials:**
   Choose suitable oils or fats (such as coconut oil, palm oil, or stearic acid), alkali (usually potassium hydroxide for softness), water, and additives like glycerin, fragrance, and color.
2. **Preparation of alkali solution:**
   Dissolve potassium hydroxide (KOH) in distilled water carefully to prepare the lye solution.
3. **Heating of oils/fats:**
   Heat the selected oils or fatty acids to a suitable temperature (around 60–70°C) until they are completely melted.
4. **Saponification process:**
   Slowly add the alkali solution to the heated oils with continuous stirring. This leads to the formation of soap through the saponification reaction.
5. **Addition of glycerin and additives:**
   Add glycerin to improve moisturizing properties. Then incorporate other ingredients such as fragrances, colors, and preservatives.
6. **Mixing and homogenization:**
   Stir thoroughly to obtain a smooth, creamy, and uniform consistency.
7. **Cooling:**
   Allow the mixture to cool gradually. During cooling, it thickens into a soft cream-like texture.

- **Packaging:**

Transfer the prepared cream soap into suitable containers and seal properly. **(Each point have 0.75 marks)**

18. Explain the role of essential oils in improving therapeutic and sensory properties of oral care products.

1. Therapeutic roles:
Antimicrobial activity: Essential oils like clove oil (eugenol), peppermint oil, and tea tree oil help inhibit the growth of bacteria responsible for plaque, tooth decay, and gum diseases.
Anti-inflammatory effect: They reduce gum inflammation, swelling, and irritation.
Analgesic property: Some oils (e.g., clove oil) provide mild pain relief in toothaches and gum discomfort.
Antioxidant action: Help protect oral tissues from oxidative stress and promote overall oral health.
2. Sensory roles:
Flavoring agent: Provide a pleasant taste, making oral care products more acceptable and refreshing.
Aroma enhancement: Impart a fresh and clean smell, improving user experience.
Cooling sensation: Oils like peppermint give a cooling and soothing effect in the mouth.

- Mouth freshness: Help in controlling bad breath (halitosis). **(Each point have 2.5 marks)**

19. Describe the different water sampling methods used for testing water quality?

Proper sampling is essential to obtain accurate results in water analysis. The main methods include:
1. Grab Sampling:
A single sample is collected at a specific place and time. It represents the water quality at that particular moment. This method is simple and commonly used for routine analysis.
2. Composite Sampling:
Samples are collected at different times or locations and then mixed together. It provides an average representation of water quality over a period or area, useful for monitoring fluctuations.
3. Random Sampling:
Samples are taken from random locations or at random times. This helps to avoid bias and gives a general idea of overall water quality.
4. Depth Sampling:

Samples are collected from different depths of a water body (surface, middle, bottom) to study variations in water quality with depth.

5. Continuous Sampling:
Water is collected continuously over a period using automatic samplers. This method is useful for detecting changes over time, especially in industrial or polluted areas.

- 6. Sterile Sampling (for microbiological analysis):
  Samples are collected in sterile containers to prevent contamination, especially when testing for microorganisms. **(any five point have one mark)**

20. **Describe the practical procedure for analysis of soil pH and explain how the results are used in agriculture.**

Practical procedure for analysis of soil pH:

1. Sample collection:
   Collect soil samples from different spots of the field (about 10–15 cm depth). Remove stones, roots, and debris.
2. Sample preparation:
   Air-dry the soil, grind gently, and sieve to obtain a uniform sample.
3. Preparation of soil suspension:
   Take about 10 g of soil in a beaker and add 25 mL of distilled water (soil:water ratio ≈ 1:2.5). Stir well.
4. Settling:
   Allow the mixture to stand for 20–30 minutes so that soil particles settle.
5. pH measurement:
   - Use a pH meter: Calibrate with standard buffer solutions, then dip the electrode into the supernatant and record the pH.
     (or)
   - Use pH paper/indicator: Dip into the solution and compare the color with the standard chart.
6. Recording results:
   Note the pH value and classify the soil as acidic, neutral, or alkaline.

   (Each point have one mark)